



2025. Fall.

8. For continuous functions  $g: [0,1] \rightarrow \mathbb{R}$  and  $h: [-1,1] \rightarrow \mathbb{R}$ , define

$$f: [0,1] \rightarrow \mathbb{R}; x \mapsto \int_0^1 h(x-y)g(y) dy$$

For each  $n \in \mathbb{N}$ , define

$$f_n: [0,1] \rightarrow \mathbb{R}; x \mapsto \frac{1}{n} \cdot \sum_{i=0}^{n-1} h(x - \frac{i}{n}) \cdot g(\frac{i}{n})$$

Prove that  $f_n \rightarrow f$  uniformly on  $[0,1]$

Proof. Define  $\ell: [0,1] \times [0,1]; (x,y) \mapsto h(x-y) \cdot g(y)$ .

Since  $h$  and  $g$  both are continuous, therefore  $\ell$  is continuous and since  $[0,1]^2$  is compact set, so  $\ell$  is continuous uniformly.

Given  $\varepsilon > 0$ , there exists  $\delta > 0$  such that for any  $(x_1, y_1), (x_2, y_2) \in [0,1]^2$ ,

$$\|(x_1, y_1) - (x_2, y_2)\| < \delta \Rightarrow \|\ell(x_1, y_1) - \ell(x_2, y_2)\| < \varepsilon.$$

Now, given  $x \in [0,1]$ , and  $n > \frac{1}{\delta}$ , ( $n > \frac{1}{\delta}$  gives  $\frac{1}{n} < \delta$ )

$$\begin{aligned} |f(x) - f_n(x)| &= \left| \int_0^1 \ell(x, y) dy - f_n(x) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \ell(x, y) dy - \frac{1}{n} \cdot \sum_{i=0}^{n-1} \ell(x, \frac{i}{n}) \right| \\ &= \left| \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \ell(x, y) dy - \sum_{i=1}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \ell(x, \frac{i}{n}) dy \right| \\ &\leq \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \left| \ell(x, y) - \ell(x, \frac{i}{n}) \right| dy \\ &< \sum_{i=0}^{n-1} \int_{\frac{i}{n}}^{\frac{i+1}{n}} \varepsilon dy = \varepsilon. \end{aligned}$$

Therefore,  $n > \frac{1}{\delta} \Rightarrow \sup_{x \in [0,1]} |f_n(x) - f(x)| < \varepsilon$ , So, proved.  $\square$

2022 Spring.

8. Define  $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \sum_{n=1}^{\infty} \frac{1}{x^2 + n^2}$

Then,  $f$  is differentiable on  $\mathbb{R}$ .

Proof. For each  $m \in \mathbb{N}$ , define

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

At first,  $f_m(x) \rightarrow f(x)$  as  $m \rightarrow \infty$ , because that

$$\sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1 \text{ as } m \rightarrow \infty,$$

so the increasing bounded sequence  $f_m(x)$  converges. Now,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

if  $|x| \leq 1$ , then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq \frac{2}{n^4} \text{ and } \sum \frac{1}{n^4} < \infty,$$

if  $|x| > 1$ , then

$$\left| \frac{-2x}{(x^2 + n^2)^2} \right| \leq 2 \cdot \frac{|x|}{x^2 + n^2} \cdot \frac{1}{x^2 + n^2} \leq 2 \cdot \frac{1}{n^2} \text{ and } \sum \frac{1}{n^2} < \infty$$

therefore, the Weierstrass M-test gives  $f'_m$  converges uniformly.

Now,  $f_m$  converges at some point and  $f'_m$  converges uniformly on  $\mathbb{R}$ , the limit function  $f$  is differentiable.  $\square$  sm.

First, for any  $x \in \mathbb{R}$ ,  $|x^2 + n^2| \geq n^2$ , so

$$\sum_{n=1}^{\infty} \left| \frac{1}{x^2 + n^2} \right| \leq \sum_{n=1}^{\infty} \frac{1}{n^2} < \infty, \text{ because } \sum_{n=2}^m \frac{1}{n^2} < \sum_{n=2}^m \frac{1}{n^2 - n} = \sum_{n=2}^m \frac{1}{n-1} - \frac{1}{n} = 1 - \frac{1}{m} \rightarrow 1,$$

as  $m \rightarrow \infty$ . Therefore, given  $\varepsilon_0$ , for some  $N \in \mathbb{N}$ ,  $m > n \geq N$  implies

$$\sup_{x \in \mathbb{R}} \left| \sum_{k=n}^m \frac{1}{x^2 + k^2} \right| \leq \sup_{x \in \mathbb{R}} \sum_{k=n}^m \left| \frac{1}{x^2 + k^2} \right| < \sum_{k=n}^m \frac{1}{k^2} < \varepsilon.$$

Now, define for each  $m \in \mathbb{N}$ ,

$$f_m(x) = \sum_{n=1}^m \frac{1}{x^2 + n^2}.$$

We already shown  $f_m \rightarrow f$  uniformly. And,

$$f'_m(x) = \sum_{n=1}^m -\frac{2x}{(x^2 + n^2)^2}$$

converges uniformly, because

2024, Spring.

1. Let  $A, B \in M_{m \times n}(F)$ . Show that  $\text{rank}(A+B) \leq \text{rank } A + \text{rank } B$ .

[5m]

Proof. By definition,

$$\text{rank}(A+B) = \dim \text{Im}(A+B) \text{ and } \text{rank } A = \dim \text{Im } A, \text{ rank } B = \dim \text{Im } B.$$

Given  $\alpha \in \text{Im}(A+B)$ ,  $\alpha = (A+B)\beta = A\beta + B\beta \in \text{Im } A + \text{Im } B$  for some  $\beta \in F^n$ , therefore  $\text{Im}(A+B) \subseteq \text{Im } A + \text{Im } B$ . Now,

$$\text{rank}(A+B) \leq \dim(\text{Im } A + \text{Im } B) \leq \dim \text{Im } A + \dim \text{Im } B = \text{rank } A + \text{rank } B.$$

2. Suppose that  $P$  and  $Q$  are positive definite such that  $P^2 = Q^2$ . Show  $P = Q$ .

Proof. Since  $P$  is Hermitian, there exists an orthonormal basis  $\mathcal{B} = \{\alpha_1, \dots, \alpha_n\}$  such that  $P\alpha_i = C_i\alpha_i$  for some  $C_i > 0$ , for each  $i$ .

Let  $1 \leq i \leq n$  be fixed. Then, by the assumption,

[5m]

$$P^2\alpha_i = C_i^2\alpha_i = Q^2\alpha_i \Rightarrow (Q^2 - C_i^2 I)\alpha_i = (Q - C_i I)(Q + C_i I)\alpha_i = 0.$$

Since  $Q$  is positive definite,  $(Q - C_i I)\alpha_i = 0$ , because: if it is not zero, then  $(Q - C_i I)(Q + C_i I)\alpha_i = (Q + C_i I)(Q - C_i I)\alpha_i = 0$ ,

so  $-C_i$  is negative eigenvalue of  $Q$ , it contradicts to  $Q$  is positive definite. Therefore, for each  $1 \leq i \leq n$ ,

$$P\alpha_i = C_i\alpha_i = Q\alpha_i,$$

and since  $\mathcal{B}$  is basis, therefore  $P = Q$ .  $\square$

2024, Spring

3. Let  $T: \mathbb{R}^5 \rightarrow \mathbb{R}^5; x \mapsto Ax$  where

$$A = \begin{pmatrix} 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix}$$

a). There exists  $T$ -invariant subspace  $V \subseteq \mathbb{R}^5$  s.t.  $\dim V = 4$ ?

b). "

s.t.  $\dim V = 2$ ,  $T|_V$  is diagonalizable?

c). "

$\dim V = 3$ , "

Proof.

a). Let  $V = \langle e_1, e_2, e_3, e_4 \rangle$ . Then,

$$Ae_1 = e_1, Ae_2 = e_1 + e_2, Ae_3 = e_3, Ae_4 = e_3 + e_4$$

are all contained in  $V$ , so  $T[V] \subseteq V$ . (clearly  $\dim V = 4$ )

b). Let  $V = \langle e_1, e_3 \rangle$ . Then  $Ae_1 = e_1$ ,  $Ae_3 = e_3$  are contained in  $V$ , and

$$[T|_V]_{\{e_1, e_3\}} = I_2. \text{ So } T|_V \text{ is diagonalizable.}$$

c). Observe characteristic polynomial of  $T$  is  $(t-1)^5$ .

If such a subspace  $V$  exists, then the characteristic polynomial of  $T|_V$  is

$(t-1)^3$ . Since  $T|_V$  is diagonalizable if and only the minimal polynomial

splits and separable, thus the minimal polynomial of  $T|_V$  must be  $(t-1)$ .

That is,  $T|_V = I_V$  but it implies that  $\ker(T-I)$  has dimension more than or equal 3, it contradicts to

$$\dim \ker(A-I) = \dim \ker \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{pmatrix} = 2.$$

2025, Fall

1. a). Show that  $\phi: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p'(0)$  is linear functional.

8m

b). Find a dual basis corresponding to  $\{1, t, \dots, t^n\}$  of  $P_n(\mathbb{R})$ .

Proof. Let  $f(x) = \sum_{i=0}^n a_i x^i$  and  $g(x) = \sum_{i=0}^n b_i x^i$  and  $C \in \mathbb{R}$  be given.

Then,  $\phi(Cf + g) = (Cf + g)'|_{t=0} = C \cdot f'|_{t=0} + g'|_{t=0} = C \cdot \phi(f) + \phi(g)$

by the linearity of differentiation. This shows the a),

To show b), let

$$\mathcal{B}^* := \left\{ \frac{1}{k!} \cdot \phi_k: P_n(\mathbb{R}) \rightarrow \mathbb{R}: p(t) \mapsto p^{(k)}(0) \mid k=0, 1, \dots, n \right\}$$

Then, for each  $k=0, 1, \dots, n$ ,

$$\frac{1}{k!} \cdot \phi_k(t^k) = \frac{k!}{k!} \cdot 1 = 1, \text{ and } \frac{1}{k!} \cdot \phi_k(t^r) = 0 \text{ if } r \neq k.$$

Similarly a), the elements in  $\mathcal{B}^*$  are all linear functional, so the  $\mathcal{B}^*$  is dual basis.

2025. Fall

2. Let  $A \in M_{\text{max}}(1\mathbb{R})$ .

15m:  $A^n$ 의 rank,  $n=2$ 일 때부터 시작

a). For  $n \in \mathbb{N}$ ,  $\text{rank } A^n \leq \text{rank } A$ .

b). Let  $L_A: \mathbb{R}^m \rightarrow \mathbb{R}^m, x \mapsto Ax$ .

Show that for  $n \geq 2$ ,

$$\text{rank } A^n = \text{rank } A \text{ implies } \ker L_A \cap \text{Im } L_A = \{0\}.$$

Sol) a). Given  $x \in \text{Im } A^n$ ,  $x = A^n y$  for some  $y \in \mathbb{R}^m$ .

So,  $x = A^n y = A^{n-1} A y \in \text{Im } A$ , for  $n \geq 2$ . Thus  $\text{Im } A^n \subseteq \text{Im } A$ , it follows  $\text{rank } A^n \leq \text{rank } A$ . If  $n=1$ , clear.

b). Given  $x \in \ker A$ ,  $A^n x = A^{n-1} A x = A^{n-1} 0 = 0$ , so  $x \in \ker A^n$ , so  $\ker A \subseteq \ker A^n$ . Since  $\text{rank } A^n = \text{rank } A$ , the dimension theorem gives  $\text{nullity } A^n = \text{nullity } A$ .

Therefore, since  $\mathbb{R}^m$  is finite dimensional, so is  $\ker A^n$ , thus  $\ker A = \ker A^n$ .

Now, let  $x \in \ker L_A \cap \text{Im } L_A$  be given. Then,  $Ax = 0$  and  $x = Ay$  for some  $y \in \mathbb{R}^m$ .

Then,  $A^{n-1} x = A^n y = 0$ , because  $Ax = 0$ , so  $A^{n-2} Ax = 0$ , and this implies  $y \in \ker A^n$ .

Since  $\ker A^n = \ker A$ ,  $Ay = 0$ , so  $x = Ay = 0$ , so proved.  $\square$

2025. Fall

3. Find all  $a, b, c \in \mathbb{R}$  such that

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$$A = \begin{bmatrix} 2 & 0 & 0 \\ a & 2 & 0 \\ b & c & -1 \end{bmatrix} \text{ is diagonalizable.}$$

Sol. Since the characteristic polynomial of  $A$  is regardless of  $a, b, c$ ,

$$\chi_A(t) = (t-2)^2(t+1).$$

Since a Matrix is diagonalizable if and only if the minimal polynomial splits and separable, and the minimal polynomial divides the characteristic polynomial, the  $A$  is diagonalizable if and only if

$$(A-2I)(A+I) = \begin{bmatrix} 0 & 0 & 0 \\ a & 0 & 0 \\ b & c & -3 \end{bmatrix} \cdot \begin{bmatrix} 3 & 0 & 0 \\ a & 3 & 0 \\ b & c & 0 \end{bmatrix} = 0.$$

$$\text{Calculating gives } = \begin{bmatrix} 0 & 0 & 0 \\ 3a & 0 & 0 \\ ac & 0 & 0 \end{bmatrix}$$

therefore,  $a=0$ ,  $b \in \mathbb{R}$ ,  $c \in \mathbb{R}$  are all possible  $(a, b, c)$ .



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4, Let  $N \subseteq GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$  be the set of all pairs  $(A, B)$  such that  $\det A = \pm \det B$ .

20m

Show that  $N$  is normal and determine the factor group  $(G \times G)/N$ .

Sol), Let  $(A, B) \in N$  and  $(C, D) \in GL_n(\mathbb{R}) \times GL_n(\mathbb{R})$  be given. Then

$$(C, D)(A, B)(C, D)^{-1} = (CAC^{-1}, DBD^{-1})$$

And, since the determinant preserves the multiplication of matrix,

$$\det CAC^{-1} = \det C \cdot \det C^{-1} \cdot \det A = \det A$$

$$\text{and } \det DBD^{-1} = \det D \cdot \det D^{-1} \cdot \det B = \det B$$

So, since  $(A, B) \in N$ ,  $\det CAC^{-1} = \det A = \pm \det B = \pm \det DBD^{-1}$

therefore  $(C, D)(A, B)(C, D)^{-1} \in N$ . And, to show  $N$  is subgroup,

first,  $(I, I) \in N$  trivially, so  $N \neq \emptyset$ . Let  $(A, B), (C, D) \in N$  be given. Then

$$(A, B) \cdot (C, D)^{-1} = (AC^{-1}, BD^{-1})$$

satisfies  $\det AC^{-1} = \det A \cdot \det C^{-1} = \pm \det B \cdot \det C^{-1} = \pm \det B \cdot \det D^{-1} = \pm \det BD^{-1}$ .

therefore, by the subgroup criterion,  $N$  is subgroup, and normal.  $\perp$  hom.

Define a map

$$\mathcal{Q}: GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) \rightarrow \mathbb{R}^{\times}; (A, B) \mapsto (\det A - \det B^{-1})^2$$

Then,  $\ker \mathcal{Q} = N$  and  $\mathcal{Q}$  is homomorphism, onto  $(0, \infty)$ , because

the det preserves multiplication, so is  $\mathcal{Q}$ , and onto is given by: fix  $B = I$ , and  $A = \text{diag}(r, 1, \dots, 1)$  where  $r > 0$  is arbitrary.

By the first isomorphism theorem,

$$GL_n(\mathbb{R}) \times GL_n(\mathbb{R}) / N \cong ((0, \infty), \times) \xrightarrow{\perp} \text{hom.} \quad \mathcal{Q}$$

Comment:  $\mathcal{Q}$  은 먼저  $\mathbb{R}^{\times}$  구조 시작하고, hom 임을 보려면 앞 계산을 꼭 해야 함!

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10m

5. Let  $G$  be an order 21 subgroup of  $\mathbb{C}^\times$ .

a)  $G$  is cyclic.

b). For any  $g \in G$ ,  $x^{12} = g^{15}$  has 3 distinct solutions in  $G$ .

Proof. a). Since  $\mathbb{C}$  is abelian, so  $G$  is also abelian.

By the Cauchy's theorem,  $G$  has order 3 element  $x$  and order 7 element  $y$ .  
Now, since 3 and 7 are relatively prime,  $|xy| = 21$ , so  $G = \langle xy \rangle$ .

b). By a),  $G \cong \mathbb{Z}/21\mathbb{Z}$ . So, we can write the given equation as

$$12x \equiv 15g \pmod{21} \quad (g \in \mathbb{Z}/21\mathbb{Z}).$$

Regardless of  $g$ ,  $\gcd(12, 21) = 3$  divides  $15g$ , so it has three distinct solutions.

Note:  $ax \equiv b \pmod{n}$  has a solution iff  $\gcd(a, n) \mid b$ .

And in such a case, it has  $\gcd(a, n)$  solutions.

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6. Let  $\{a_n\}$  be a bounded sequence in  $\mathbb{R}$ . Define

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$$\{a_n\}^* = \{x \in \mathbb{R} \mid x \text{ is a subsequential limit of } \{a_n\}\}$$

a). Prove that  $\{a_n\}^*$  is closed.

b). Show  $\limsup_{n \rightarrow \infty} a_n = \sup \{a_n\}^*$

c). Is there a sequence  $\{b_n\}$  such that  $\{b_n\}^* = [0, 1]$ ?

Proof a). If  $\{a_n\}^*$  is finite, then clearly closed in  $\mathbb{R}$ .

Assume that  $\{a_n\}^*$  is infinite. Since  $\{a_n\}$  is bounded, so the set of subsequential limits  $\{a_n\}^*$  is bounded set. So it has a limit point.

Let  $\alpha$  be a limit point of  $\{a_n\}^*$ . Then, given  $\epsilon \in \mathbb{R}$ , there exists a

subsequential limit  $p$  of  $\{a_n\}$  such that  $p \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$ .

For  $k=1$ , choose such a  $p_1 \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$ , and since  $p$  is a subsequential limit, choose  $a_{p_1} \in B_{\frac{1}{k}}(p)$ , then  $a_{p_1} \in B_{\frac{\epsilon}{2}}(\alpha)$ .

For chosen  $a_{p_1}, \dots, a_{p_{k-1}}$ , choose  $p_k \in B_{\frac{\epsilon}{2}}(\alpha) \setminus \{\alpha\}$ .

Then, similarly, there is  $a_{p_k} \in B_{\frac{\epsilon}{2}}(\alpha)$  such that  $p_k > p_{k-1}$ .

because no such  $p_k$  exists, then it is contradiction to  $a_{p_k}$  is a subsequential limit. Therefore, the arbitrary limit point  $\alpha$  is contained in  $\{a_n\}^*$ ,

so it is closed.

b). Since a),  $\{a_n\}^*$  is closed and bounded, so compact. Thus

$$\sup \{a_n\}^* = \max \{a_n\}^* = \alpha \text{ for some subsequential limit } \alpha.$$

So, given  $\epsilon > 0$ , there exists  $N \in \mathbb{N}$  such that  $n \geq N \Rightarrow a_n < \alpha + \epsilon$  (since  $\alpha = \max \{a_n\}^*$ )

and there exists infinitely many terms in  $\{a_n\}$  such that  $\alpha - \epsilon < a_n$  (since  $a_{p_k} \rightarrow \alpha$ )

Thus  $\limsup_{n \rightarrow \infty} a_n = \alpha$

c). Let  $b_n$  be defined as:

$$\frac{1}{1}, \frac{1}{2}, \frac{2}{2}, \frac{1}{3}, \frac{2}{3}, \frac{3}{3}, \dots$$

Then, given  $b \in [0, 1]$ , there exists a subsequence in  $\{b_n\}$  converges to  $b$ , because: for any  $\epsilon > 0$ , there are infinitely many terms in  $b_n$  such that

$$|b_n - b| < \epsilon,$$

so we can construct the subsequence. (20)

2025, Fall.

7. Let  $E$  be a nonempty bounded subset of a Metric space  $(X, d)$ .

10m

Define  $\text{diam } E := \sup\{d(x, y) \mid x, y \in E\}$

a). Prove  $\text{diam } E = \text{diam } \bar{E}$

b). If  $E$  is compact, then  $d(x, y) = \text{diam } E$  for some  $x, y \in E$ .

Proof.

a). Since  $E \subseteq \bar{E}$ ,  $\text{diam } E \leq \text{diam } \bar{E}$  is clear.

To show the inverse inequality, using the property of supremum.

Given  $\varepsilon > 0$ , there exists  $x, y \in \bar{E}$  such that

$$\text{diam } \bar{E} - \frac{\varepsilon}{2} < d(x, y) \leq \text{diam } \bar{E}$$

And, since  $\bar{E}$  is closure, there are  $x', y' \in E$  such that

$$d(x', x) < \frac{\varepsilon}{4} \quad \text{and} \quad d(y, y') < \frac{\varepsilon}{4}.$$

Now,  $\text{diam } \bar{E} < d(x, y) + \frac{\varepsilon}{2} < d(x, x') + d(x', y') + d(y', y) + \frac{\varepsilon}{2} < d(x', y') + \varepsilon$ .

Since  $\varepsilon > 0$  was arbitrary,  $\text{diam } \bar{E} \leq \text{diam } E$ , so proved. 7m.

b). Since  $E$  is compact set, so  $E \times E$  is also compact set.

Since the metric function  $d$  is continuous, it preserves compactness, so  $d[E \times E] \subseteq \mathbb{R}$  is compact set. Therefore,

$$\text{diam } E = \max d[E \times E] \in d[E \times E],$$

so there is  $(x', y') \in E \times E$  such that  $d(x', y') = \max d[E \times E]$  3m.

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8. Let  $a > 1$ . Evaluate

5m

$$\lim_{t \rightarrow 0^+} \int_t^{at} \frac{\cos x}{x} dx$$

Proof. Consider the Taylor expansion

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}$$

Therefore  $\frac{\cos x}{x} = \frac{1}{x} - \frac{x}{2!} + \frac{x^3}{4!} - \dots$

Now, 
$$\int_t^{at} \frac{\cos x}{x} dx = \int_t^{at} \frac{1}{x} + \int_t^{at} -\frac{x}{2!} + \frac{x^3}{4!} - \dots dx$$
$$= \ln a + \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx$$

(It is possible, because  $\sum (-1)^n x^{2n-1}/(2n)!$  converges for all  $x \in \mathbb{R}$ )

Now, 
$$\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} \int_t^{at} (-1)^n \frac{x^{2n-1}}{(2n)!} dx = 0,$$
 so

the given limit is  $\ln a$ .

Revised in next

Observation for  $\lim_{t \rightarrow 0^+} \int_t^a \frac{\cos x}{x} dx$ .

As we know, the power series representation

$$\frac{1}{x} \cdot \cos x = \frac{1}{x} \cdot \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!}$$

is valid for all  $x \neq 0$ . Therefore, for any  $t > 0$ ,

$$\begin{aligned} \int_t^a \frac{\cos x}{x} dx &= \int_t^a \frac{1}{x} + \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} dx \\ &= \int_t^a \frac{1}{x} dx + \int_t^a \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} dx \end{aligned}$$

Meanwhile, the series converges uniformly in  $[t, a]$  for small enough  $t > 0$ , specifically,  $t < \frac{1}{a}$ , then for each  $n \in \mathbb{N}$ ,

$$\left| (-1)^n \frac{x^{2n-1}}{(2n)!} \right| \leq \left| \frac{1}{(2n)!} \right| \quad \text{and} \quad \sum_{n=1}^{\infty} \frac{1}{(2n)!} < \infty,$$

therefore the Weierstrass M-test gives the condition,

Thus, we can exchange the integral to the sum,

$$\int_t^a \sum_{n=1}^{\infty} (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} \int_t^a (-1)^n \frac{x^{2n-1}}{(2n)!} = \sum_{n=1}^{\infty} (-1)^n \left( \frac{1}{2n \cdot (2n)!} \cdot (a^{2n} t^{2n} - t^{2n}) \right)$$

Converges uniformly similarly, now, we can exchange the limit to the sum

$$\begin{aligned} &\lim_{t \rightarrow 0^+} \sum_{n=1}^{\infty} (-1)^n \frac{1}{2n \cdot (2n)!} \cdot t^{2n} (a^{2n} - 1) \\ &= \sum_{n=1}^{\infty} \lim_{t \rightarrow 0^+} \quad // \\ &= \sum_{n=1}^{\infty} 0 \\ &= 0 \end{aligned}$$

Consequently, the given integral is

$$\int_t^a \frac{1}{x} dx = \ln a - \ln t = \ln a.$$

Note:  $\frac{\cos x}{x} = \frac{1}{x} + \frac{\cos x - 1}{x}$ ,  $\left| \int_t^a \frac{\cos x - 1}{x} dx \right| \leq \int_t^a \frac{x}{2} dx \rightarrow 0$  as  $t \rightarrow 0$ .

$\Gamma KLG$ , 73p.

2.7.14.

If  $\{x_n\} \subset B$  converges to  $x \in B$ , then

$$\bigcap_{k=1}^{\infty} \{x_n \mid n \geq k\} = \{x\}.$$

$\Gamma_{KKG}$ ,

2.7.7, and 2.7.20.

Let  $A, B \subseteq \mathbb{R}^n$ .

1) If  $A$  is open, then  $A+B$  is open.

2) If  $A, B$



TSNU. Q. 14.

Prove that  $\lim_{b \rightarrow \infty} \int_0^b \frac{\sin x}{x} dx$  exists.

Proof. First, let  $a_n = n\pi$  ( $n \in \mathbb{N}$ ). Then, a sequence  $\left\{ \int_0^{a_n} \frac{\sin x}{x} dx \right\}$  converges, because:

$\left| \int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx \right|$  is decreasing and converges to zero, and

$\int_{n\pi}^{(n+1)\pi} \frac{\sin x}{x} dx$  is positive if  $n$  is even and is negative if  $n$  is odd.

That is,  $\int_0^{n\pi} \frac{\sin x}{x} dx = \sum_{k=1}^n \int_{a_{k-1}}^{a_k} \frac{\sin x}{x} dx$  is alternating series, so converges.

12.26. Fall.

8. For  $f \in C([0,1])$ , define its extension

$$\tilde{f}(x) := \begin{cases} f(0) & \text{if } x < 0 \\ f(x) & \text{if } 0 \leq x \leq 1 \\ f(1) & \text{if } x > 1 \end{cases}$$

For each  $n \geq 1$ , define

$$p_n(x) := \frac{n}{2} \cdot \chi_{[-\frac{1}{n}, \frac{1}{n}]}(x)$$

Define  $T_n f(x) := \int_{\mathbb{R}} p_n(x-y) \tilde{f}(y) dy$ ,  $x \in \mathbb{R}$ .

a). Show that  $T_n f \rightarrow f$  on  $[0,1]$ .

b). Determine  $T_n f \rightarrow f$  uniformly on  $[0,1]$  or not.

Proof. Given  $x \in \mathbb{R}$  and  $n \in \mathbb{N}$ ,  $x-y \in [-\frac{1}{n}, \frac{1}{n}] \Leftrightarrow -\frac{1}{n} + x \leq y \leq \frac{1}{n} + x$ . Therefore,

$$T_n f(x) = \int_{-\frac{1}{n}+x}^{\frac{1}{n}+x} \frac{n}{2} \cdot \tilde{f}(y) dy.$$

Since  $\tilde{f}$  is continuous on  $[-1,2]$ , so it is continuously uniformly.

Given  $\varepsilon > 0$ , take  $\delta > 0$  such that  $\forall x, t \in [0,1]$ ,  $|x-t| < \delta \Rightarrow |f(x) - f(t)| < \varepsilon$ .

Now, given  $x \in [0,1]$ , and  $n > \frac{1}{\delta}$ ,

$$\begin{aligned} |T_n f(x) - f(x)| &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} \tilde{f}(y) dy - \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} f(x) dy \right| \\ &= \left| \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} \frac{n}{2} [\tilde{f}(y) - f(x)] dy \right| \\ &\leq \frac{n}{2} \cdot \int_{x-\frac{1}{n}}^{x+\frac{1}{n}} |\tilde{f}(y) - f(x)| dy \\ &< \frac{n}{2} \cdot \frac{2}{n} \cdot \varepsilon = \varepsilon \end{aligned}$$

Therefore,  $\sup_{x \in [0,1]} |T_n f - f| \rightarrow 0$  as  $n \rightarrow \infty$ , so converges uniformly.